



Orion-MC

Manual Ventilation

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1 Introduction

The Orion-MC is the communication interface between a horticulture computer (the main controller) and a variety of systems in the horticultural greenhouse (ventilation).

The horticulture computer communicates with the Orion-MC using separate control signals.

The Orion-MC communicates with the IO modules via a CAN Local data bus. The Orion-MC receives control commands from the main controller. The Orion-MC interprets these control commands in terms of commands for the relevant IO module. The Orion-MC also sends the main controller feedback on the position adopted by the controlled element.

The Orion-MC also offers a number of additional functions. You can view the information transmitted by EC fans on the Orion-MC.



Safety instructions and warnings

This product should be installed by an approved, certified installer in accordance with the relevant standards (BS, ANSI, ISO, for example).

The warranty is void if this product is installed in a switchbox other than by Hotraco Horti BV. The installation incorporating this product should be equipped with an appropriate safety system that is inspected at periodic intervals (at least once a day) to verify that it is working correctly.

Please note that when the product is included in a climate control system, the oxygen content in the climate controlled area must be guaranteed at all times, including during an emergency. To guarantee the oxygen content we recommend an emergency system, which must be periodically inspected for proper operation.

An emergency system can consist of:

- an emergency power generator;
- emergency opening systems;
- provision for manual operation.

We wish to draw your attention to the following:

- The Orion-MC computer is an electronic product. Consequently, you always need to take into consideration a possible malfunction.
The alarm contacts of the various computers should be connected to a central alarm system. Furthermore, Hotraco Horti BV recommends the installation of an additional independent alarm system (such as a min./max. thermostat).
Although Hotraco Horti BV has made every effort to ensure that adequate alarms are generated in the event of emergencies, Hotraco Horti BV cannot guarantee that all possible situations have been covered.
- We recommend that the computer remains powered up whenever possible. It is preferable to keep the computer powered up when not in use, since this will prevent condensation.
- Do not clean the computer with running water: the computer is splashproof, but not waterproof!
- This appliance is covered by the warranty and liability regulations laid down in documents including the Hotraco Horti general terms of sale governing the contract concluded for the delivery of this appliance.
- Use other than described in this manual is not permitted without express consent from Hotraco Horti BV.
- The Orion-MC is under no circumstances a safety component.
- The Orion-MC may only be operated and adjusted by persons who are familiar with the operating instructions provided in this manual.
- The person who operates or adjusts the Orion-MC must be aware of all potential dangers of the systems that the Orion-MC controls.
- A defective Orion-MC is hazardous! In the event of a fault, the systems in which the Orion-MC is used are to be taken out of service until the Orion-MC has been repaired.

- It is not permitted to connect external devices that do not have a SELV or Class II power supply (double insulated).
- Hotraco Horti wishes to point out that Machinery Directive 2006/42/EC is applicable when the Orion-MC is used to control moving systems.
- Due to the nature of the Orion-MC controllers, a controlled system can start at any time. Hotraco Horti would like to point out that this may result in hazardous situations and danger from moving parts when such a system is in operation. Therefore, people are not permitted to be in the immediate vicinity of moving parts. The moving parts must be fitted with guards and/or safety devices.
- Motors and actuators that are directly controlled via the Orion-MC may never be connected without a suitable motor protector and a lockable isolating switch.

End of life (WEEE)

The crossed-out wheeled bin indicates a product that, at the end of its life cycle, must be collected separately (Directive 2012/19/EU on waste electrical and electronic equipment - WEEE). Separate collection is a precondition to ensure specific treatment and recycling of WEEE to get the chosen level of protection of human health and the environment in the Union. The black bar below the crossed-out wheeled bin indicates that the product was put on the market on or after August 13, 2005



The European Union has systems in place for the separate collection of waste electrical and electronic products and batteries. Thus, return waste products, batteries and accumulators to the appropriate local authorities, or a local collection point/recycler. When you do not dispose of the product correctly, you may be fined.

2 Operation

2.1 Operating keys

The Orion-MC is equipped with a number of operating keys. These keys can be classified into the following groups:

Function keys:  Time and date, groups, fans



 No function

 Alarms

 Diagnostics

 Installation procedure

Navigation keys:  Move cursor

 Increase value / Previous line

 Decrease value / Next line

 Move cursor / Correction of numeric entry

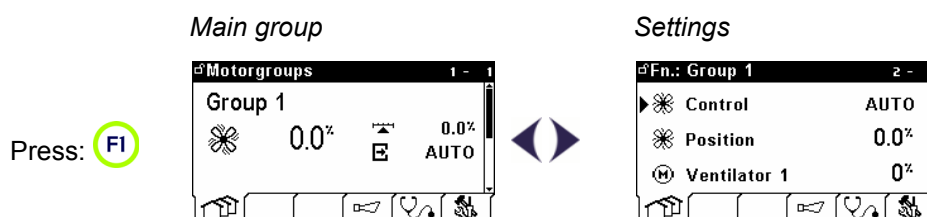
 Confirm value

 Next motor or motor group

 Previous motor or motor group

2.2 The readout

The display can show information in several ways. The section describes the main ways.



The *main group readout* displays the current data and/or the status of the controlled element. These data relate to the measured and calculated positions and can be fetched by pressing the **F1** key.

The data in the *main group readout* cannot be changed. You can change settings relating to the data displayed in the main group by pressing the **▶** key, which opens the settings readout. Press the **◀** key to return to the *main group readout*.

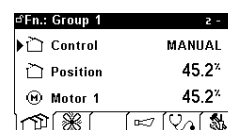
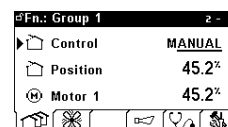
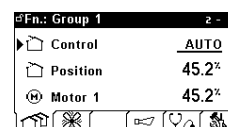
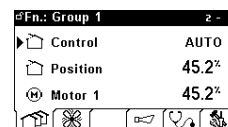
A list of the possible main groups and settings is enclosed in appendix 1. Data relating to controlled elements that have not been activated are skipped automatically and, consequently, cannot be fetched.

2.3 Entering and adjusting settings

Press the (**F1** or **F2**) function key to display the *main group screen*. Pressing the **▶** key opens the *settings screen*. You can fetch the required setting using the **▲** and **▼** keys.

Example Changing a group to manual operation mode:

1. Press **F1** or **F2**: the main group is displayed:
2. Press **▶**: the settings screen is displayed.
3. Press **▶**: the cursor is moved to the setting.
4. Press **▲**: the setting is changed.
5. Press **OK**: the changed setting has been confirmed.



2.4 System settings

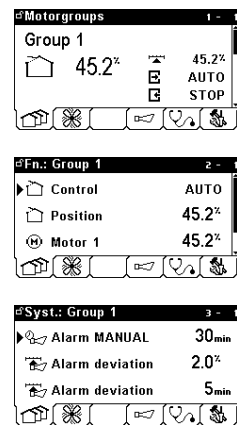
System settings refer to settings relating to the minor changes that can be made to the system. The basic settings have already been configured: the Orion-MC will carry out its control functions satisfactorily without any changes to the system settings.

However, you can change the system settings to tailor the Orion-MC precisely to the equipment that is to be controlled. We recommend that you make any changes in consultation with your installer.

The system settings are grouped by controlled element. The system settings for the relevant controlled element can be fetched and adjusted by opening the *settings screen* and pressing the **F6** key. The system settings that are not of relevance are not displayed.

Example System settings of a group:

1. Press **F1** or **F2**: the main group is displayed:
2. Press **▶**: the settings screen is displayed.
3. Press **F6**: the system settings are displayed. System settings are adjusted using the procedure used for the adjustment of settings, see chapter 2.3.

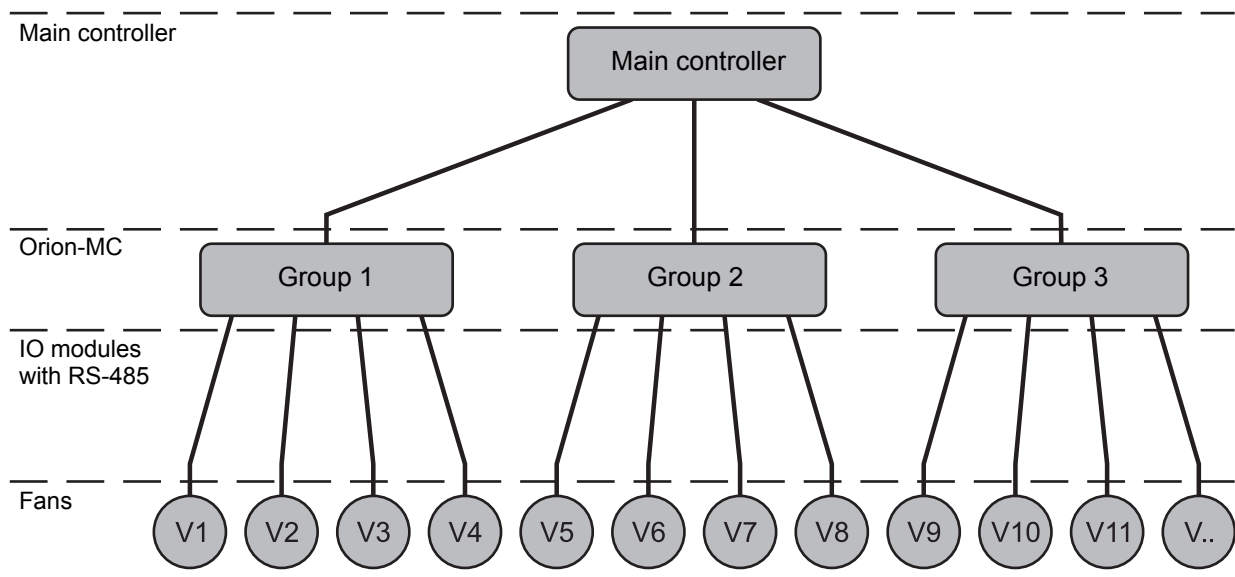


3 The Orion-MC's function

3.1 Control of the ventilation systems

The building is equipped with a number of fans (V1, V2, etc., in the following diagram). These fans are connected to Hotraco Horti IO modules via RS-485 communication. The IO modules are in turn connected to the Orion-MC. More information is enclosed in Appendix .

The fans are clustered into fan groups in the Orion-MC. Each fan group consists of one or more fans that are, in principle, controlled simultaneously and all adopt the same position. The main controller controls the fan groups and the Orion-MC ensures that the fans adopt the correct position.



You can use the Orion-MC to control all the fans in a group or, if you so wish, to control a single fan.

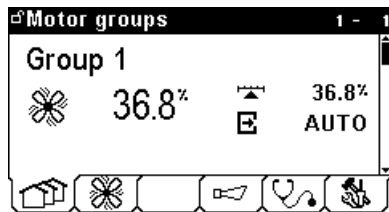
3.2 Virtual control

The horticulture computer serves as the controller and determines the required action for each group. The horticulture computer sends the relevant action to the Orion-MC. The Orion-MC simulates this action and sends the horticulture computer feedback on the simulated action (= virtual control). The horticulture computer interprets this feedback as confirmation that the required action has been carried out. At the same time the Orion-MC controls the IO modules by sending commands to ensure that the group carries out the required action. The relevant units send feedback on the positions to the Orion-MC. The action is stopped once the required position has been reached or an alarm is generated if the required position is not reached.

4 Ventilation settings

4.1 Ventilation groups

Press **FI** to display the main **group** screen. You can use the **▲▼** keys to browse through the groups.



This screen displays the status of a **group** with fans.

36.8% = The average ventilation position of all fans in the group. This can differ from the required position (setpoint).

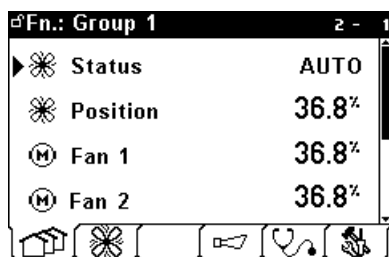
▲ = The required ventilation position for the group, calculated by the main controller.

E = The status of the group in the Orion-MC:

- AUTO** The group is set to automatic operation mode. The main controller determines the required position and the Orion-MC ensures that the fans in the group adopt the required ventilation position.
- MANUAL** The group is set to manual operation mode. The group can be controlled via the settings in the Orion-MC.
- OFF** The group is switched off: the ventilation position is 0%.

4.2 Fan group settings

Press **►** in the main group readout to display the settings readout. Navigate through the groups with **◀▶**.

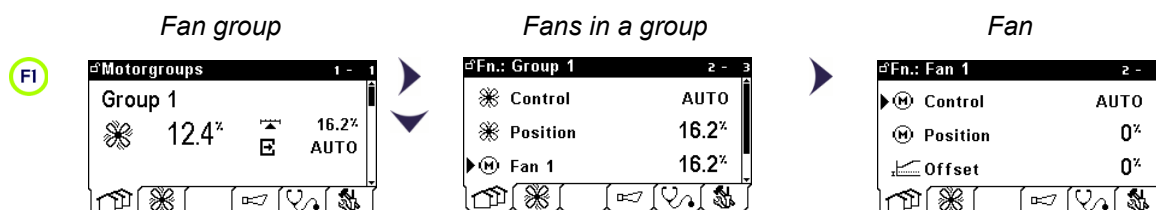


This screen displays the settings of the **fan group**.

- ✱ = *Operation*: Change the status of the group:
 - AUTO* The group is controlled automatically; the main controller determines the required ventilation position and the Orion-MC ensures that the fans rotate at the correct speed.
 - MANUAL* The group is controlled manually; the group can be controlled (using the *position* setting) from the Orion-MC.
 - OFF* The group is switched off: the ventilation position is 0%.
- ✱ = *Position*: The required ventilation position for the group, calculated by the main controller. When the *Operation* of the group is set to *MANUAL* then you can set the ventilation position here.
- Ⓜ = *Fan ...*: The current ventilation position of each fan. This example is of a group with two fans. Press ► to view the **fan** settings (see chapter 4.3).

4.3 Fan settings

The settings of a single **fan** are fetched as follows.



You can use the ◀► keys to browse through the fans. The screen displays the status of a single fan.

- Ⓜ = *Operation*: The operation mode of the fan can be set as follows:
 - AUTO* The fan is set to automatic operation mode. The Orion-MC makes sure that the fan rotates at the correct speed.
 - MANUAL* The fan is set to manual operation mode. The required ventilation position can be set under *Position*.
 - OFF* The fan is switched off: the ventilation position is 0%.
- Ⓜ = *Position*: The fan's ventilation position. When the *Operation* of the fan is set to *MANUAL* then you can set the ventilation position here.



= **Offset**: It is possible that a fan in a group needs to be set to a higher or lower ventilation position than the other fans. If necessary, you can set an incremental increase or decrease in the ventilation position of each fan (the *offset*). The *offset* is enabled solely in the automatic operation mode. **Example**

Fan	Offset	Required position	Actual position
No. 1	10%	70%	80%
No. 2	0%	70%	70%
No. 3	-8%	70%	62%
No. 4	0%	70%	70%
No. 5	8%	70%	78%

4.4 Ventilation group system settings

In the **ventilation group** settings screen, press **F6** to fetch the system settings. You can use the **◀▶** keys to browse through the ventilation groups.

SYS Description

	Alarm, manual
	Alarm variance (min. variance before alarm is generated)
	Alarm variance (min. wait time before alarm is generated)
	Alarm, urgent
	Minimum
	Maximum

Default	User
30 min	min
2%	%
5 min	min
1x	x
0%	%
100%	%

Alarm, manual

When the **ventilation group** is still set to manual operation mode on the Orion-MC after the set time then the group's alarm contact is de-energised. This function can be disabled by setting *alarm, manual* at 0 minutes.

Alarm variance

You can configure the Orion-MC to generate an alarm in the event that a fan is not at the required ventilation position after the set *wait time*.

Alarm, urgent

You can make a distinction between the alarms generated for a group of fans, namely between an alarm generated by one fan (less urgent) and an alarm generated by a number of fans (more urgent). Two alarm contacts are then assigned. The urgent alarm contact is de-energised only when the number of fans generating an alarm is greater than the number set under *alarm, urgent*.

Example

A system is equipped with 140 fans: 10% of these fans can fail without causing disruption to the process. *alarm, urgent* should then be set to 14x: the urgent alarm contact is de-energised only once more than 14 fans generate an alarm.

Minimum and maximum

The *minimum* and *maximum* settings can be used to limit the fans' ventilation position range. When the required ventilation position is below the *minimum* then the ventilation position is equal to the *minimum* setting. When the required ventilation position is above the *maximum* then the ventilation position is equal to the *maximum* setting. These limits are also applicable to fans that fall outside these limits as a result of the *offset* settings for the fans (see chapter 4.3). till set to manual operation mode then the group's alarm contact is de-energised. This function can be disabled by setting *alarm, manual* at 0 minutes.

You can configure the Orion-MC to generate an alarm when a flap motor is not at the required position after the set *wait time*.

5 Alarms

5.1 Alarm messages

5.1.1 Alarm types

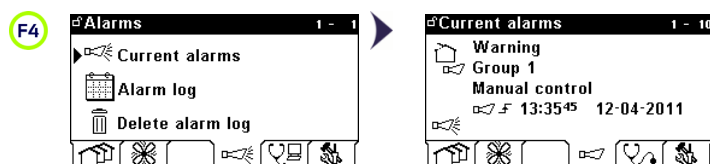
The Orion-MC uses the following types of alarms (F4).

-  = Soft alarm. The alarm contacts on the IO boards remain energised.
-  = Several soft alarms.
-  = Hard alarm. The alarm contacts on the IO boards are de-energised.
-  = Several alarms, including at least one hard alarm.

Each IO board has a separate alarm contact that is de-energised in the event of a communication failure, power failure and a **hard alarm**. The alarm contact activates, for example, a claxon, revolving lamp and/or telephone dialler. Your installer will configure the necessary settings for additional alarm contacts during the installation procedure. The following options are available:

- Each group has a separate alarm contact that is de-energised in the event of a **hard alarm** relating to the relevant group.
- A separate alarm contact that is de-energised on each **soft alarm**.

5.1.2 Alarm display



5.1.3 Resolve the alarms

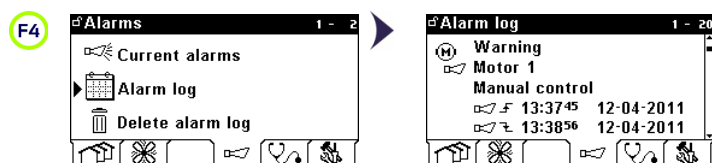
Use the following procedure in the event of an alarm:

- Press F4 to view which alarm has been generated. When there are several alarms then you can use the   keys to browse through the alarms.

2. Most alarms can be converted into another type of alarm. You do so by pressing **OK** for the alarm you wish to convert and then using the **▼▲** keys to change the type of alarm.
 -  = **Soft alarm:** The alarm has yet to be resolved, but the alarm contacts on the IO boards are re-energised. The alarm contact of the group remains de-energised. The separate alarm contact for soft alarm is de-energised.
 -  = **Suppressed soft alarm:** The alarm has not yet been resolved but will no longer be generated. The alarm contact of the group remains de-energised. When the alarm contact for soft alarm is de-energised then this contact is re-energised.
 -  = **Hard alarm:** When the alarm contact for the group has been de-energised then this remains de-energised. When the alarm contact for soft alarm is de-energised then this contact is re-energised. The alarm contacts on the IO boards are de-energised.
 -  = **Deleting an alarm because it has been resolved:** If the alarm has been resolved then you can delete the alarm to restore the control.
3. Investigate the cause of the alarm.
4. Solve the problem in a **safe manner**.
5. Many alarm messages will be cancelled automatically once the problem has been resolved. When necessary, alarms will be cancelled only once you have deleted the message with the  command. When an alarm cannot be cancelled then there is still a problem.
6. Once all alarms have been cancelled *No current Alarms* is displayed in the screen.

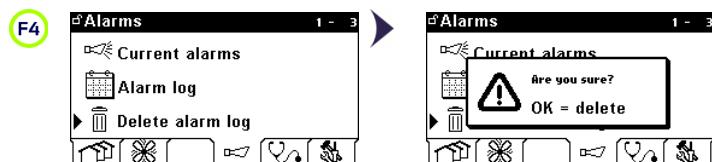
5.1.4 Alarm log

Once an alarm has been resolved then the alarm message is saved in *Alarm log*. The time at which the alarm was resolved is added to the last line. You can use the keys to browse through the alarm messages.



5.1.5 Deleting alarms

You can use *Delete alarm log* to delete the alarms.



5.2 Alarm messages from the IO modules

5.2.1 Messages via RS485 (ebmBus or Modbus)

The control board of a single fan can send the following alarm messages. Consult the fan documentation to resolve the problem. Once the problem has been resolved then the alarm can be deleted in the Orion-MC:

1. Call the alarm in *current alarms* (see also chapter 5.1.3).
2. Press **OK** once you see the alarm displayed on the screen.
3. Delete the alarm with **OK** .

When there is more than one alarm message then you can delete all the alarm messages using the *Reset all* diagnostics function, see chapter 6.2.

5.3 Orion-MC Alarms

Description	Type	Cause
 Fan 'xx' Manual mode.	Alarm/Warning	Fan number 'xx' is set to manual operation mode (in the Orion-MC). The alarm contact will be de-energised after a while (see system settings).
 Group 'xx' Manual mode.	Alarm/Warning	The group number 'xx' is set to manual-operation mode in the Orion-MC. The alarm contact will be de-energised after a while (see system settings).
 Fan 'xx' Position not reached.	Alarm	Fan number 'xx' has a different speed, (see).

5.4 System alarms

5.4.1 General alarms

Description	Type	Cause
 Orion switched off	Warning	The software has switched off the computer at the specified time.

The computer has been switched off and gives an alarm. Press any key to suppress the alarm and to convert the alarm into a warning. No alarms from the controllers are shown while the computer is switched off.

Description	Type	Cause
 System restart	Warning	The computer has been restarted at the specified time.
 Settings deleted	Alarm	The settings have been deleted at the specified time (Installation procedure).
 Options deleted	Alarm	The options have been deleted at the specified time (Installation procedure).
 Alarm deleted	Warning	The alarms have been deleted at the specified time (see chapter 5.1).
 Options restored	Alarm	Someone has restored the options.
 Settings restored	Alarm	Someone has restored the settings.
 Options + settings restored	Alarm	Someone has restored the options and settings.
 Options (+ settings) not restored	Alarm	The options and settings that have been restored are not correct.

5.4.2 Communication alarms

Description	Type	Cause
 Board IO-xx-xx [xx] not found.	Alarm	The Orion has no connection with the relevant IO board. Possible causes: • Communication interrupted. • Broken contact. • CAN bus terminator defect.
 No alarm contact.	Alarm	No alarm contact assigned. The IO boards are not configured correctly (Installation procedure).
 Fan 'xx' not found.	Alarm	There is a communication problem with the RS485 connection (ebmBus, Modbus on the IO-07-07 board).

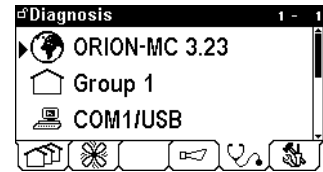
5.4.3 System alarms (internal)

The Orion-MC tests a number of internal functions that are directly applicable to the computer. On detecting an error the computer generates a system alarm displaying one of the following messages. Always warn your installer in the event of a system alarm!

Description	Type	Cause
 Options corrupted	Alarm	System setup no longer correct.
 Settings corrupted	Alarm	System settings are no longer correct.
 Minimum/maximum value corrupted	Warning	The stored minimum and maximum values are no longer correct.
 I ² C error	Alarm	Internal communication problem in the Orion.
 EEPROM error	Alarm	Internal communication problem with the EEPROM settings and options.
 RTC error	Warning	Problem with the real-time clock (program flow).
 TIMER error	Warning	Internal clock is not working correctly.
 HTRAP error	Warning	Internal memory is not working correctly.
 PLL error	Warning	PLL circuit is not working correctly.
 IO board ADC error	Warning	Analog/Digital converter not working properly.

6 Diagnostics

Press the **F5** diagnostics keys to get access to extra functions. You can use these functions to access information about the system elements. This information is clustered by controlled group. The diagnostic functions are displayed by pressing the **►** key for the relevant element.



You can also fetch information about the various communication lines. Only the installed communication lines are displayed (see chapter 6.3).

You can also use the diagnostics screen to read the data of a SD card or write data to a SD card. More information is given in chapter 6.4

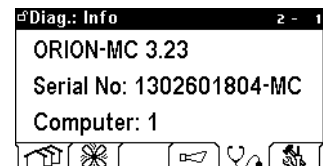
6.1 Software version, serial number and module

Software version

The diagnostics opening screen displays the software's version (in this example Orion-MC 4.18). Since the majority of this computer's functions are determined by the software it is important that the number in front of the decimal point (in this example 4.18) is in agreement with the number of this manual (see the version number on the front page + footer texts).

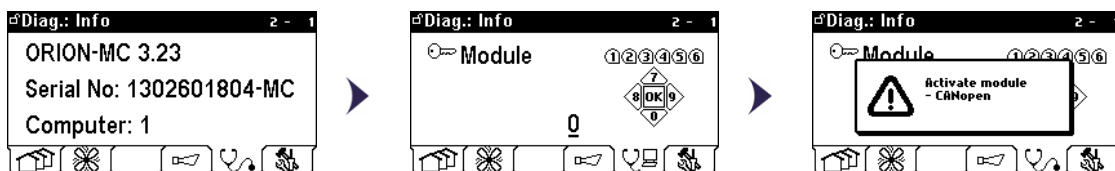
Serial number

Each Orion is assigned a unique serial number. When the cursor is on *Orion-MC*, pressing the **►** key displays the screen with the serial number. This can be applicable to a software update.



Module

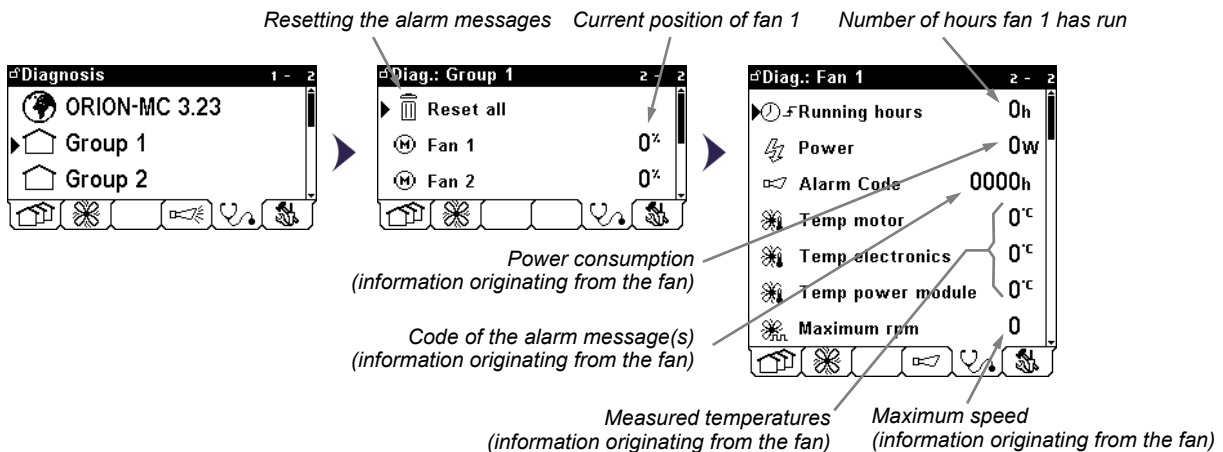
You can change or add a module (for example, CANopen communication) by entering a code under *Module* (**↩**).



Note You can request the code from your supplier.

6.2 Ventilation control

Pressing on the  key when the cursor is on *Group* fetches diagnostics information for the group with fans. You can use the   keys to select one of the fans in the group. You cannot make any changes.

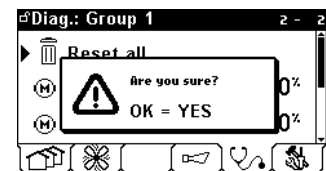


Note

- It is possible that the diagnostics information is not displayed directly when the fan does not send any information or sends other information.
- A speed of 100% indicates that the fan is rotating at maximum speed.

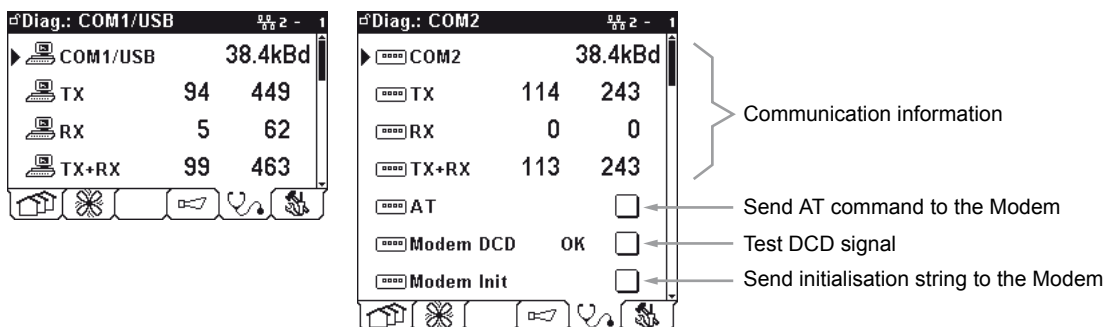
Reset all

It is possible that different alarm messages are sent to the Orion-MC when the system includes a large number of fans. Once the failure has been resolved then the alarm messages can be deleted with the *Reset all* function. Confirm the reset with the **OK** key.



6.3 Communication

6.3.1 COM1/USB and COM2



6.3.2 CAN Local (IO boards)

Diag.: CAN LOCAL		
TXOK	1	225
RXOK	1	241
TXOK+RXOK	0	459
BOFF	0	
EWRN	0	
Stuff error	0	
Form error	0	
Ack error	0	
Bit1 error	0	
Bit0 busoff error	0	
Bit0 normal error	0	
Crc error	0	

Communication information

6.4 SD card

6.4.1 SD card settings

Diagnosis 1 - 40

- CAN LOCAL
- SD-CARD**
- Display Green

Diag.: SD-CARD 2 - 1

Diag.: SD-CARD		
SD	SD-CARD	active
SD	SD-CARD	927792 991488
SD	AL.001	SIZE 120061
SD	AL.001	19.42.52 06-06-2008
SD	OPTA.001	SIZE 65908
SD	OPTA.001	11.13.40 26-05-2010
SD	SETP.001	SIZE 0
SD	SETP.001	00:00.00 00-00-0

SD card status

Files on the SD card

Status of the file

File information

Changing the SD card status

Press the **▶** key when the cursor is on *SD CARD*. Change the status with the **▲▼** keys. Confirm with the **OK** key. You can assign the following statuses to the SD card:

- Remove:** The SD card can be removed safely from the Orion-MC.
- Ask remove:** Data is still being written to the card, the SD card cannot be removed yet.
- Current:** The SD card can be used to save or restore data.

Changing the status of the file

Press the  key when the cursor is on the file. Change the status with the   keys. Confirm with the **OK** key. You can assign the following statuses to the SD card:

-  No action takes place.
-  The data are stored on the SD card. With log data the data are supplemented continually and the  symbol is always displayed. With one-off data (such as options) the symbol changes back to  once the data has been saved.
-  The data are read out from the SD card and send to the Orion (for example, options). Once the read-out has been completed the symbol changes back to .

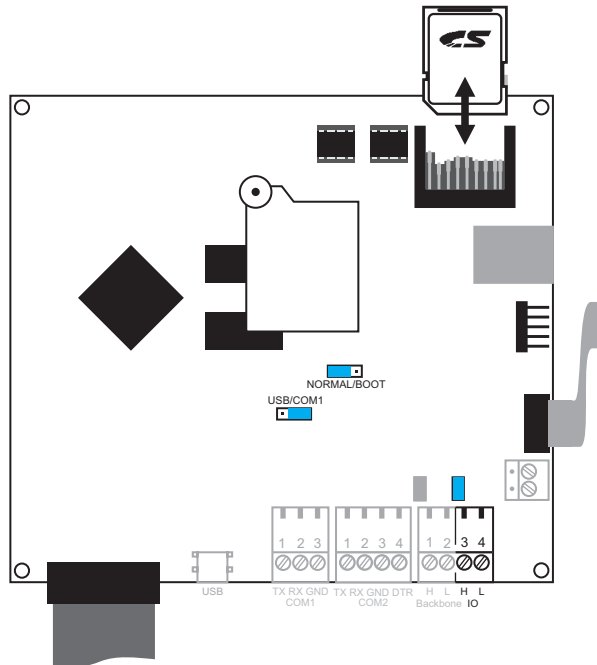
Data on the SD card

Alarm list (AL), options (OPT) and setpoints (SETP).

6.4.2 SD card location

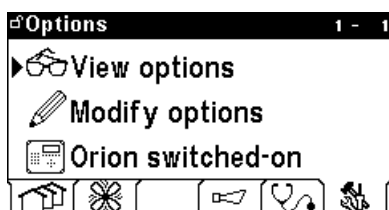
The SD card is in the case of the Orion-MC. Open the case to access the SD card. The SD card is located on the CPU board. Carefully slide the SD card into or out of the holder. Only FAT-16 or FAT-32 format SD cards can be used: SDHC cards cannot be used.

Note Make sure that the status of the SD card is set to *Remove* (see chapter 6.4.1).

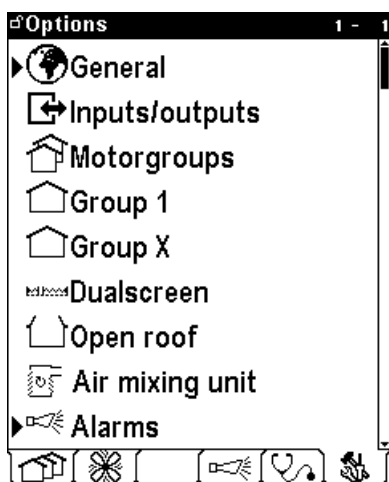


7 Installation procedure

Once the Orion-MC has been connected in accordance with the connection diagrams (see chapter 9) then you can enable the required options. Pressing the **F6** key when the main group screen is displayed opens the following screen:



This screen can be used to view and/or change the options and switch the Orion-MC on or off with the software. Moving the cursor to *View options* or *Change options* and then pressing **►** opens the option groups, provided that these are not secured with an access code. You always have access to *View options*, even when changes cannot be made without an access code. The access code is set in the *General* group (see chapter 7.1). The option groups that can be configured during the installation procedure are:



Note Your screen may differ from the screen shown above: solely the option groups of the available modules are displayed!

You can navigate through the various option groups with **▼** or **▲** and then **►**. In the options screen you can use **▼** and **▲** to navigate through all questions: press **►** to change the option settings.

Fill in the configuration of each option group in the charts shown on the following pages.

7.1 General options

The general options include the language, communication settings and access codes.



General

Options, general	
Work through options?	
N	Y
Language?	
Delete options? (*1)	
Delete setpoints?	
Brightness?	
LCD dimming?	
Computer number:	
COM1/USB Yes/No (*2)	
N	Y
COM1/USB kBd	
COM1/USB Modem Yes/No	
N	Y
ATS0=	
COM2 Yes/No (*3)	
N	Y
COM2 kBd	
COM2 Modem Yes/No	
N	Y
ATS0=	
Ethernet Yes/No (*4)	
N	Y
Port:	
IP: . . .	
Mask: . . .	
Gate: . . .	
User's code: (*5)	
Installer's code:	
PC code:	
Options OK?	
Options	

- (*1) Set the required language. If so required, you can delete existing options and setpoints. You can adjust the brightness of the screen. Activating LDC dimming will automatically dim the screen after 1 minute of no activity.
- (*2) Enable the COM1 or USB port here. In addition, set the communication speed (default: 38.4 kBd). Under Answer (ATS0=), set the number of rings before the modem answers.
- (*3) Enable the COM2 port here. In addition, set the communication speed (default: 38.4 kBd). Under Answer (ATS0=), set the number of rings before the modem answers.
- (*4) Enter the network settings here.
 Port = the port number.
 IP = the IP address of the Orion
 Mask = Definition of the local network.
 Gate = IP address of the router.
 Use the ◀ and ▶ keys to browse through the settings. You can use the ▼ and ▲ keys to change the figures.
- (*5) A password consists of digits. When the password is to be entered the keys of the Orion are assigned the numerical values displayed on the screen (F1=1, F2=2, ▲=7, etc.). Confirm the entry with **OK**. Under Repeat, enter your selected password again.
 Passwords may not begin with 0 and must consist of at least 1 and at most 4 digits.
 Entering 0 (▼) disables the password function.
 The user's access code can be used to disable access to the control settings and installation procedure. The settings are automatically disabled once you have not pressed any key for 10 minutes.
 The installer's access code can be used to secure access to the installation procedure.
 The PC code secures logging in via PC software.

7.2 IO module settings

You use the IO modules options to configure the inputs and outputs.

Example You have connected an analog control signal from the main controller to a IO-07-07 module. You now need to specify the signal received by this input in the IO options.



Options: IO

IO-07-07[1]

IO-08-09[-]

IO-EKU[-]

IO-H1MC[-]

IO-H1MC[-]

(*6) The standard Orion-MC is supplied with one integral IO-07-07. You can connect additional IO modules via CAN Local (see chapter 9.3.1).

Options: IO-07-07

Work through options?

N

Y

Number of IO module: 1..16

Analog inputs: (*7)

Analog outputs: (*8)

Digital outputs: (*9)

RS485 Yes/No (*10)

N

Y

Baud rate: kBd

Parity: Non / Even / Odd

Alarm contact Yes/No (*11)

Copy options to IO number: (*12)

Options OK?

Y

Options: IO

(*7) Enter the CAN Local address configured on the IO module here.

(*8) Press **OK** to set the minimum and maximum of the output (voltage).

(*9) Press **OK** to test the output with the **▼▲** keys.

(*10) Enable the RS485 communication (ebmBus or Modbus) of the IO module here. Set the data speed and parity to the correct values.

- ebmBus; 9.6kBd, no parity
- Modbus Ziehl-Abegg; 19.2kBd, even parity
- Modbus EBM; 19.2kBd, even parity

(*11) Yes = the alarm contact of this module is de-energised in the event of a hard alarm. No = the alarm contact is de-energised only when the CAN-Local communication is broken or in the event of a power failure.

(*12) The options of identical IO modules are often the same. You can use *Copy options* to copy the IO module's settings.

Enter the CAN Local address of the new IO module and confirm with **OK**. Confirm again with **OK**. The new IO module will then be created with the same options. You can then continue immediately with the options of the newly created IO module.

Options: H1MC/H2MC/EKU (*13)

Work through options?

N

Y

Number of IO module: (*14)

-

1..32

Motor control:

(*15)

Alarm contact

Yes/No (*16)

Options OK?

Y

Options: IO

(*13) The options for the ECU, H2MC and H1MC are virtually identical.

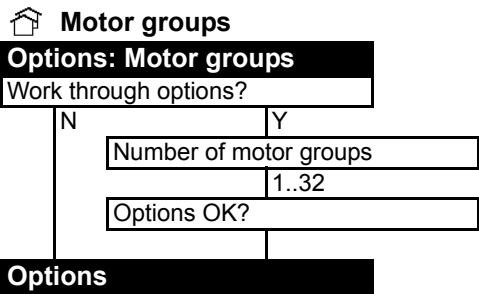
(*14) Enter the CAN Local address configured on the IO module here.

(*15) Enter the correct type of motor.
Window motor (100% = open, 0% = closed)
Screen motor (100% = open, 0% = closed)
You can also view the current setting for the position here.

(*16) Yes = the alarm contact of this module is de-energised in the event of a hard alarm. No = the alarm contact is de-energised only when the CAN-Local communication is broken or in the event of a power failure.

7.3 Motor groups

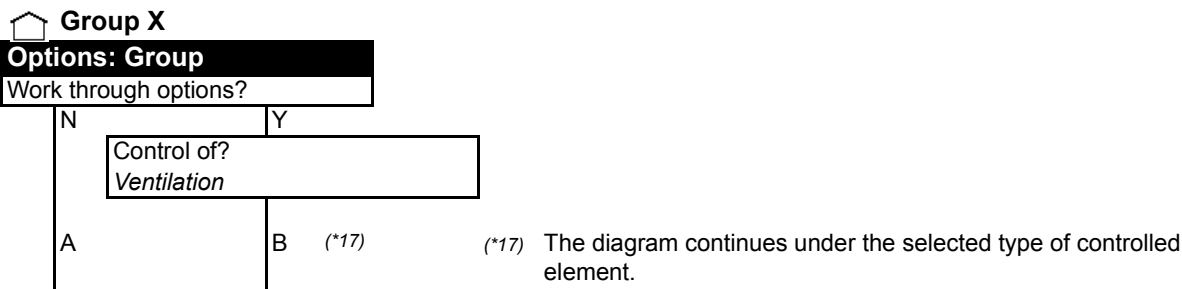
A motor group consists of motors (fans that are controlled simultaneously).
See also Appendix 2. Configure the number of motor groups here.



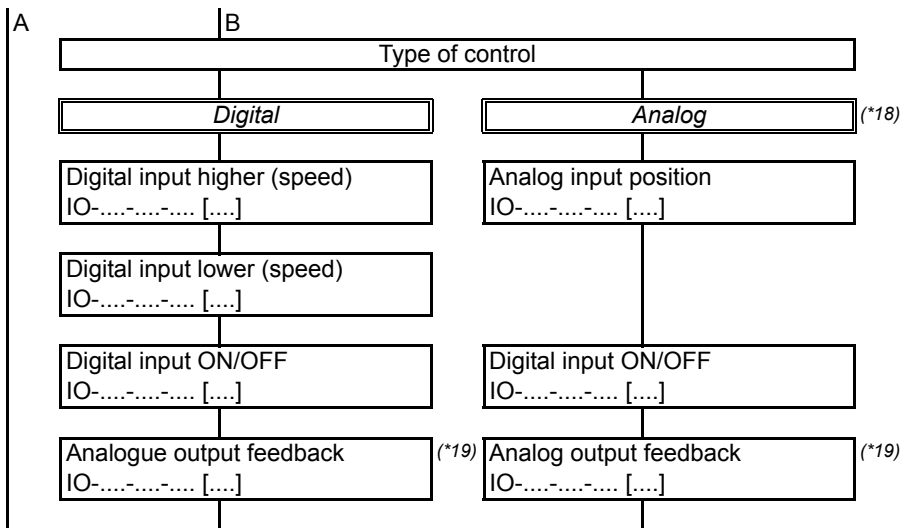
7.4 Group settings

You can configure the options for each group, namely:

- window motors (airing)
- screen motors
- fan motors



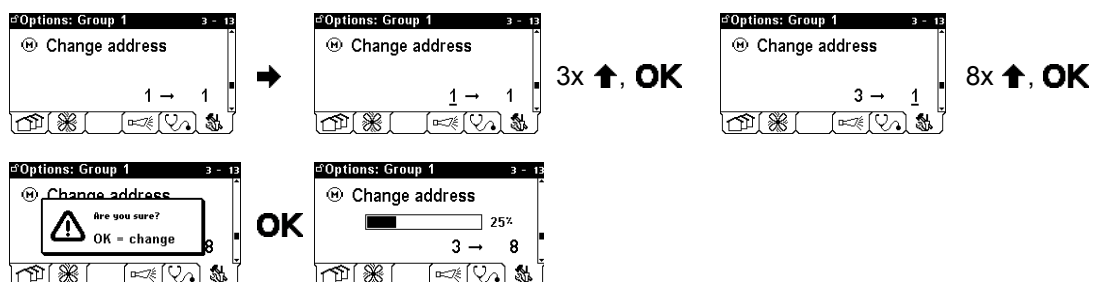
Ventilation



Running time	sec	(*20)
Number of motors:		(*21)
Communication type:	ebmBus/Modbus	
Motors	IO-...-...-... [....]	
Change address:	1 → 1	(*22)
Position	%	(*23)
Alarm contact for group	IO-...-...-... [....]	(*24)
Alarm contact, urgent	IO-...-...-... [....]	
Options OK?	Y	

Options

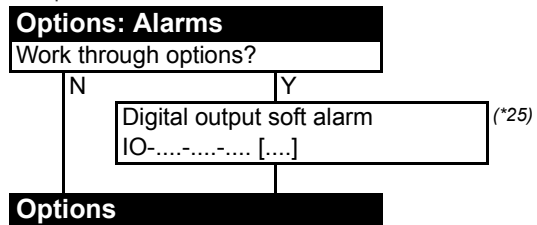
- (*18) Specify whether the control signals from the main controller are analogue or digital here. These options are not applicable to CANopen or BACnet communication.
- (*19) Specify the analogue output here when feedback to the main controller is necessary.
- (*20) Set the running time needed to control the fans from 0% to 100% here. The running time is of importance to the feedback to the main controller (the accuracy of the feedback increases with longer running times).
- (*21) Configure the number of fans in the group here, together with the communication protocol that is used (ebmBus or Modbus) and the IO modules (CAN-IO-07-07[...]) to which the relevant fans are connected. The fans are numbered sequentially. When, for example, group 1 contains 20 fans then the fans are numbered from 1 to 20. When, for example, group 2 contains 14 fans then these fans are numbered from 21 to 34. The next fan in group 3 will then be 35. See also Appendix . **Each group may not have more than 4 different RS484-busses!**
- (*22) Configure the data bus address of the fans here. For the purposes of communication with the Orion-MC the address of the fan **needs to be** equal to: *the number of the fan + 1*. **Example** fan number 1 is assigned the address 2 (1+1), fan number 14 is assigned the address 15 (14+1). See also Appendix . Example: changing address 3 to address 8.



- (*23) This option is applicable when the main controller controls the position of the ventilation on the basis of a running time. You will then need to configure the main controller's output position. **Example** The main controller calculates that the setting of the ventilation should be 45.3%. Enter 45.3% under *position*.
- (*24) You can configure two alarm contacts, namely a general alarm contact and an urgent alarm contact. The general alarm contact is de-energised in the event of a failure. The urgent alarm contact is de-energised only when a specific number of fans report a failure, see also chapter 4.4.

7.5 Alarms

Alarms



(*25) You can assign a contact to a soft alarm here. This contact will then be de-energised in the event of a soft alarm.

8 Technical specifications

8.1 Orion-UPM-30

IMPORTANT!

- When an installation has two or more components that are fed separately, check if all components can operate at the same mains voltage.
- Always replace a defective fuse with a fuse of the same brand, type and fuse rating.

Electrical

Types of mains supply	: Fixed wiring
Operating area	: Indoor, normal, dry (not suitable for wet areas)
Operating conditions	: Continuous operation
Mains power supply	: 100-240 VAC
Frequency	: 50-60 Hz
Power consumption	: 0.75-0.5 A
PCB fuse	: T800mA (5 mm x 20 mm)
Pre-fuse	: Max. 16 A
Insulation	: Class 1 (IEC 61140)
Transient over-voltage	: Category II (IEC 60664-1)
Pollution degree	: Category 2 (IEC 61010-1)

Mechanical

Mobility	: Fixed position (the product is mounted to the wall)
Operating temperature range	: 0 - 40 °C
Relative humidity	: Max. 80%
Installation height	: Max. 2000 m above sea level
Dimensions (H x W x D)	: 220 x 165 x 105 mm
Housing	: IP54, ABS
Weight	: ± 1 kg

Power supply (UPM-30)

Voltage 24 VDC $\pm 2.5\%$	: Max. 800 mA
----------------------------	---------------

Communication

CAN Local	: Max. 500 m @ 100 kbps
CAN Backbone	: Max. 500 m @ 125 kbps
RS232	: Max. 15 m @ 38400 baud
USB	: Max. 5 m @ 115 kbps
Ethernet	: Max. 100 m @ 10 Mbps

Cables

CAN Local / CAN Backbone	: Shielded twisted pair, 1x2x0.64 mm ² (CAN-BK cable)
RS232	: MSK-9
USB	: USB with USB-B mini plug and USB-A plug

Ethernet : UTP cat 5e

Compliance

EU Directives : See the Declaration of Conformity for this product

8.2 CAN-IO-07-07

IMPORTANT!

- This product can have more than one power source. Always disconnect all the power sources before you do work on live or exposed parts of the product.
- When an installation has two or more components that are fed separately, check if all components can operate at the same mains voltage.
- Always replace a defective fuse with a fuse of the same brand, type and fuse rating.
- Always prevent ground loops when you install the product. If possible, provide galvanic isolation.

Electrical

Types of mains supply	: Fixed wiring
Operating area	: Indoor, normal, dry (not suitable for wet areas)
Operating conditions	: Continuous operation
Mains power supply	: 230 VAC $\pm 10\%$
Frequency	: 50/60 Hz
Power consumption	: max. 15 VA
PCB fuse	: T100mA (5 mm x 20 mm)
Insulation	: Class 1 (IEC 61140)
Transient over-voltage	: Category II (IEC 60664-1)
Pollution degree	: Category 2 (IEC 61010-1)

Mechanical

Mobility	: Fixed position (the product is mounted to the wall)
Operating temperature range	: 0 - 40 °C
Relative humidity	: Max. 80%
Installation height	: Max. 2000 m above sea level
Dimensions (H x W x D)	: 220 x 165 x 105 mm
Housing	: IP54, ABS
Weight	: ± 1 kg

Analog inputs (4)

Temperature sensor (PT1000)	: -40 - 100 °C ± 0.5 °C
Potentiometer	: 5 k Ω
0-20 mA	: $R_i = 250 \Omega$
0-10 V	: $R_i = 50 \text{ k}\Omega$

Digital inputs (3)

Contact / counter	: NPN / PNP sensor, 24 VAC/DC, 8 mA, max. 10 Hz
-------------------	---

Analog outputs (2)

0-10 VDC	: 0-10 V, max. 1 mA
----------	---------------------

Digital outputs (4 + 1 alarm output)

Relay output K1-K4	: 0.5 A, 230 VAC
Alarm output	: 0.5 A, 24 VAC/DC

Power supply, 24 VDC

Voltage 24 VDC -30% +10%	: Max. 300 mA
--------------------------	---------------

Communication

- CAN Local : Max. 500 m @ 100 kbps
- RS485 : Max. 1200 m (depends on the speed)

Cables

- CAN Local : Shielded twisted pair, 1x2x0.64 mm² (CAN-BK cable)
- RS485 : Shielded twisted pair, 1x2x0.64 mm² (RS485 cable)

Compliance

- EU Directives : See the Declaration of Conformity for this product

9 Connection diagrams

9.1 Connection sequence

Read these instructions thoroughly before you begin work on the assembly and connections.

This product should be installed by an approved installer with certification according to the relevant prevailing standards (for example, ANSI, BS, ISO). The installer must possess the knowledge of and experience with electronic control equipment required to install the Orion-MC.

IMPORTANT! For your safety, you must work through the installation procedure in the following sequence.

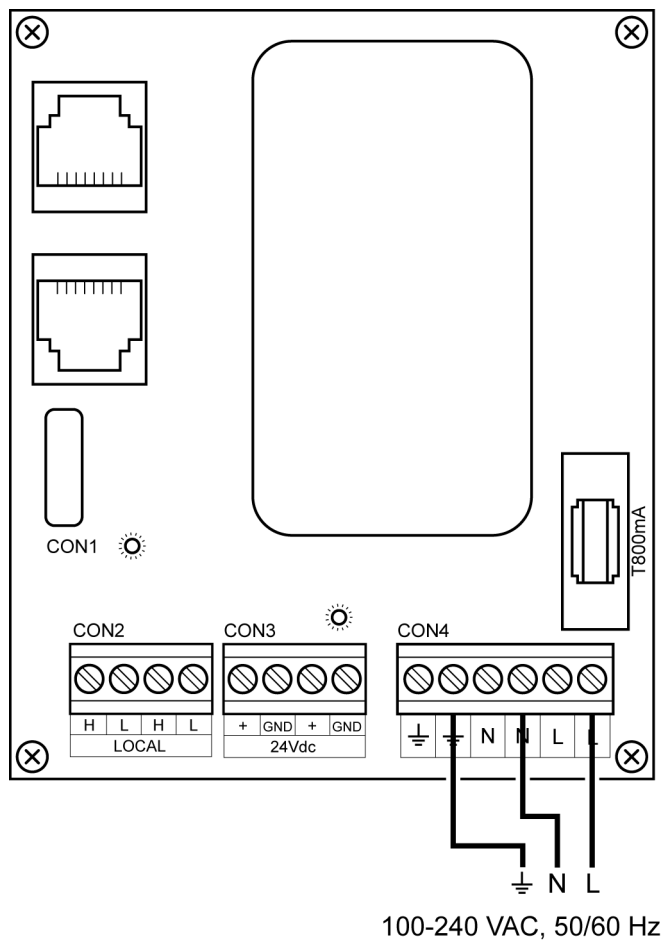
1. Install the necessary equipment (see also Appendix 2).
2. Install the necessary IO modules (see also Appendix 2).
3. Read in all motors and equipment with position feedback: these are the motors controlled using the IO modules such as the H1MC, H2MC and CAN-IO-EKU modules. Reading in these motors and equipment will ensure that they do not go out of range during the installation of the Orion-MC.
4. If possible, set all IO modules and connected equipment in manual operation mode: this will make sure that the motors and equipment to be controlled by the unit will not be started unintentionally during the installation of the Orion-MC.

Note The following connection diagrams explain the communication with the main controller, and do not extend to the connection of options (such as temperature sensors, RS485 communication, etc.). Information about the connection of options is enclosed in the relevant CAN IO module manual. In addition, see the examples of connections in Appendix 2.

9.2 Orion-UPM-30

IMPORTANT!

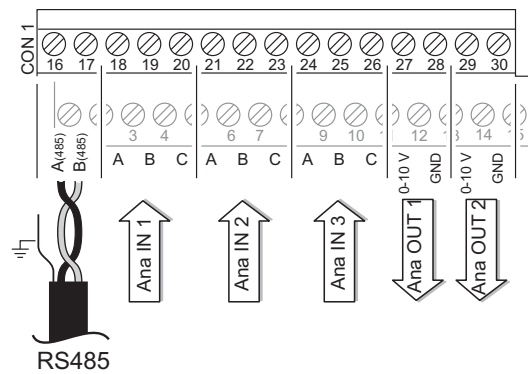
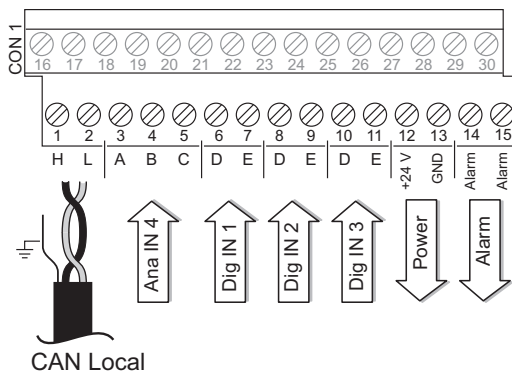
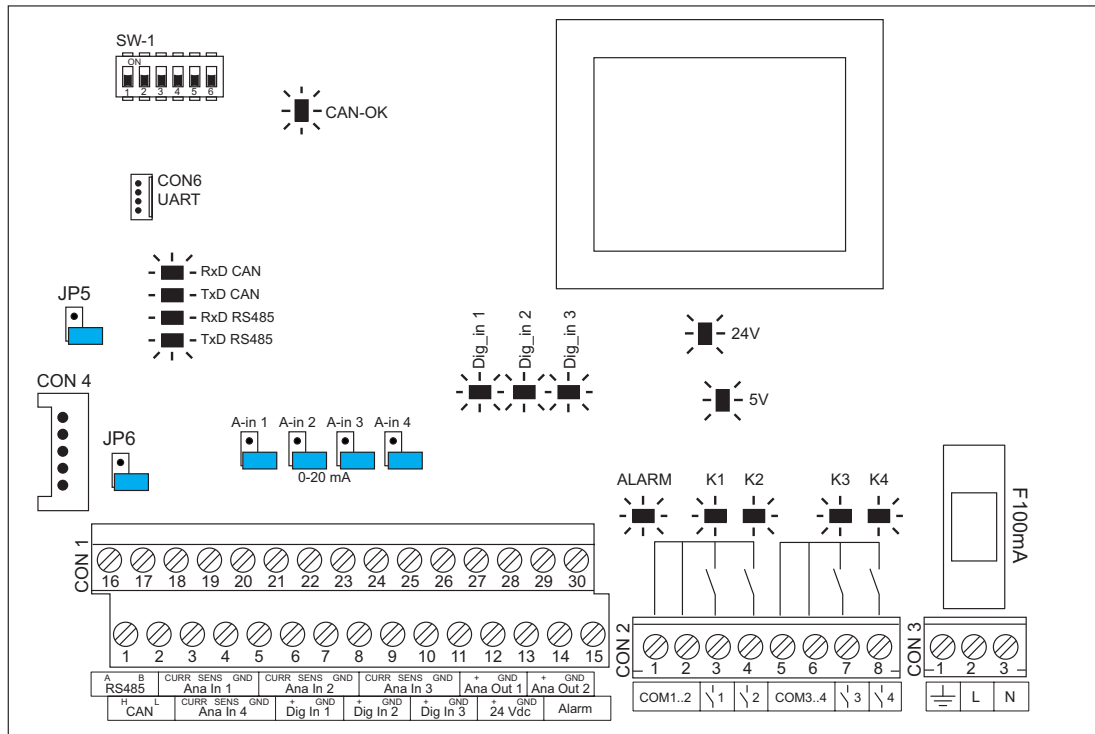
- Always disconnect the power supply to the product before you do work on electrical parts inside the product. When this is not possible, be very careful that you do not touch live parts.
- This product can have more than one power source. Always disconnect all the power sources before you do work on live or exposed parts of the product.
- When an installation has two or more components that are fed separately, do a check if all the components can operate at the same mains voltage.
- All the electrical parts inside this product are susceptible to electrostatic discharge (ESD).
 - Always disconnect the power supply to the product before you do work on electrical parts inside the product.
 - Always wear an ESD wrist strap that is connected to the main earthing system. See IEC61340-5-2 for more information on precautions with regard to ESD.
- Always prevent ground loops when you install the product. If possible, provide galvanic isolation.
- A mix of voltages at the relay contacts can damage your product.
 - Connect the digital outputs (relay contacts) to the same electrical potential.
 - Do not mix mains and low-voltage signals.



9.3 CAN-IO-07-07P

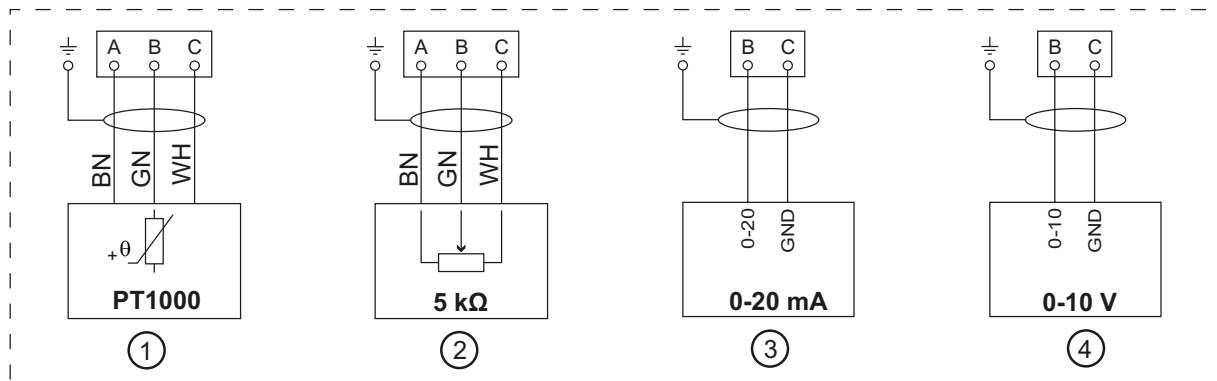
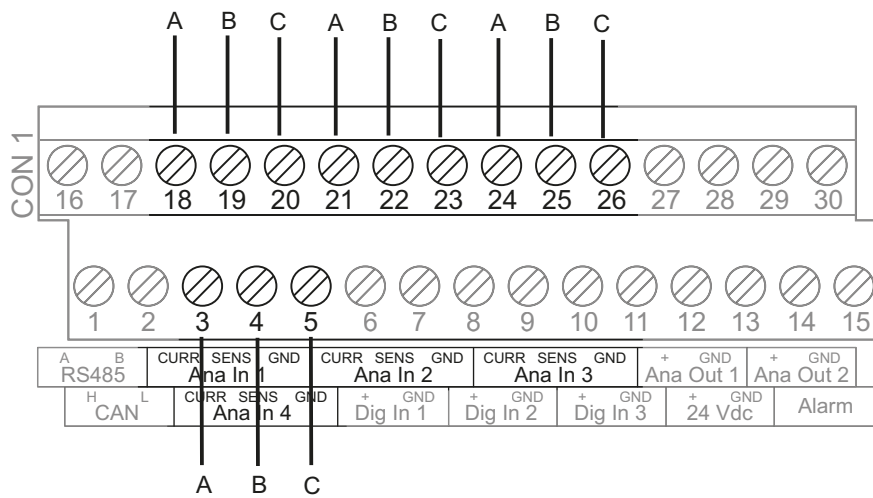
IMPORTANT!

- Always disconnect the power supply to the product before you do work on electrical parts inside the product. When this is not possible, be very careful that you do not touch live parts.
- This product can have more than one power source. Always disconnect all the power sources before you do work on live or exposed parts of the product.
- When an installation has two or more components that are fed separately, do a check if all the components can operate at the same mains voltage.
- All the electrical parts inside this product are susceptible to electrostatic discharge (ESD).
 - Always disconnect the power supply to the product before you do work on electrical parts inside the product.
 - Always wear an ESD wrist strap that is connected to the main earthing system. See IEC61340-5-2 for more information on precautions with regard to ESD.
- Always prevent ground loops when you install the product. If possible, provide galvanic isolation.
- A mix of voltages at the relay contacts can damage your product.
 - Connect the digital outputs (relay contacts) to the same electrical potential.
 - Do not mix mains and low-voltage signals.



9.3.1 Analog inputs

The analog inputs are intended for the control signals from the main controller. See Appendix 2.1 for examples of connections. The analog inputs are assigned during the installation procedure.



Legend:

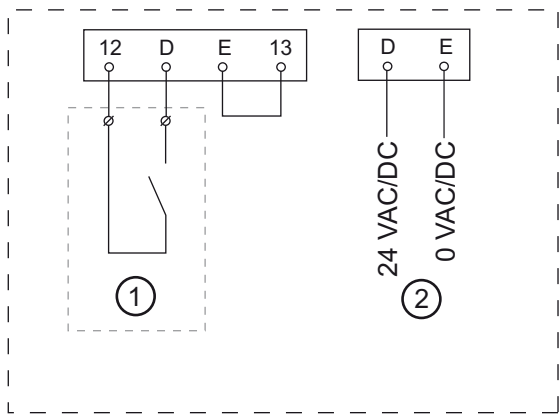
1. Temperature
2. Potentiometer
3. Analog input 0-20 mA
4. Analog input 0-10 V

Note Make sure that jumper A-in (x) for the 0-20 mA analog input is set correctly.

Note the inputs in use in the following table.

Analog input	Terminal			Selection							Description
				 Window	 Screen	 Ventilation	 Flap	 Heating	 Alarm	 Frost guard	
1	19	-	20								
2	22	-	23								
3	25	-	26								
4	4	-	5								

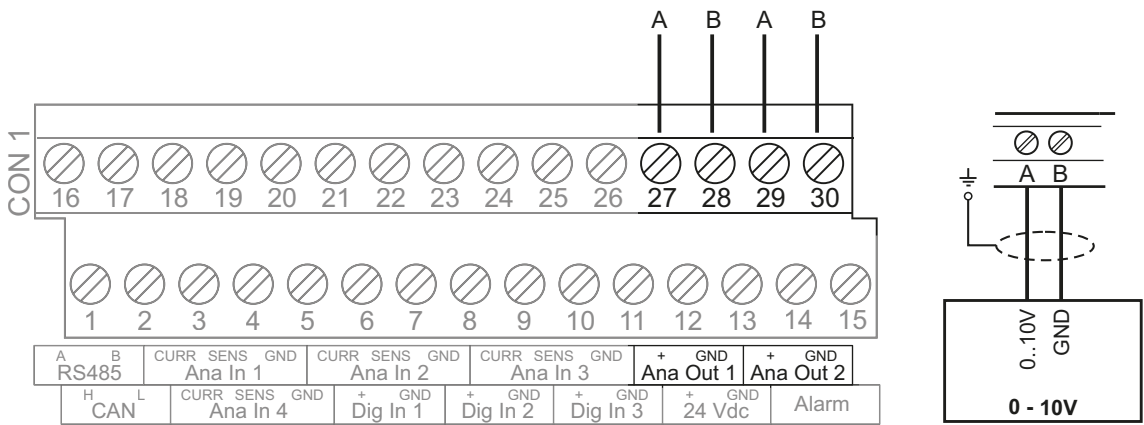
9.3.2 Digital inputs



- Legend:
- 1. External contact, for example a frost guard or motor protection
 - 2. External control signals

9.3.3 Analog outputs

The analog outputs are intended for the feedback to the main controller. See Appendix 2.1 for examples of connections. The analog outputs are assigned during the installation procedure.

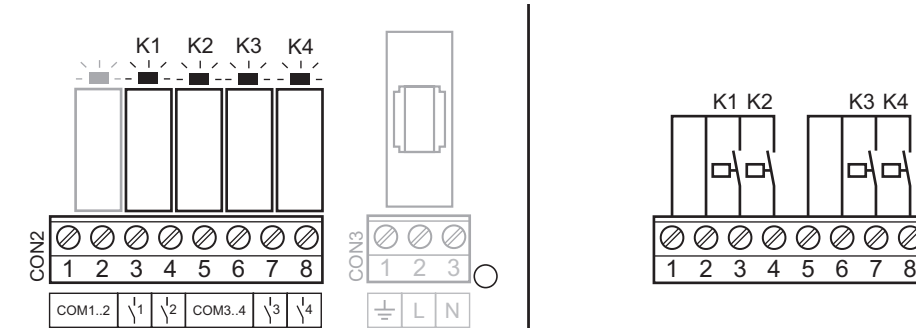


Note the outputs in use in the following table. In addition, state the range of the output (e.g, 1-9V).

Analog output	Terminal A - B	Selection:							Range Min / Max	Description
		Window position	Screen position	Fan speed	Flap position	Heater intensity				
1	27 - 28									
2	29 - 30									

9.3.4 Digital outputs

The digital outputs are intended for the feedback to the main controller. See Appendix 2.1 for examples of connections. The digital outputs are assigned during the installation procedure.

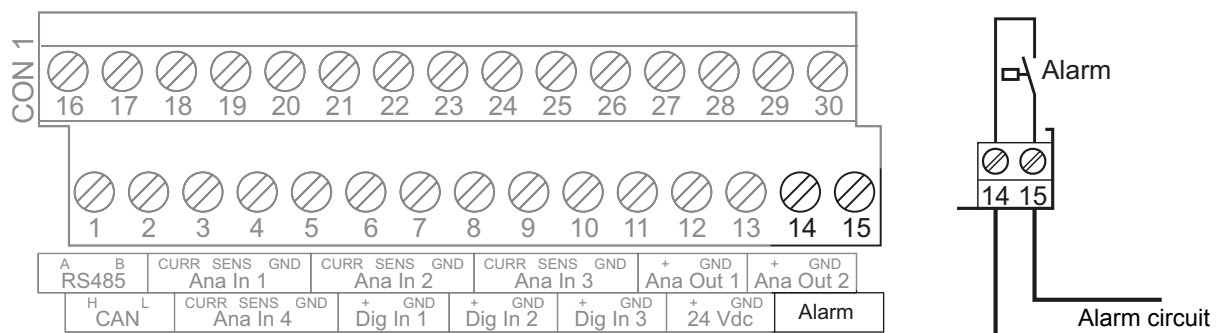


Note the outputs in use in the following table.

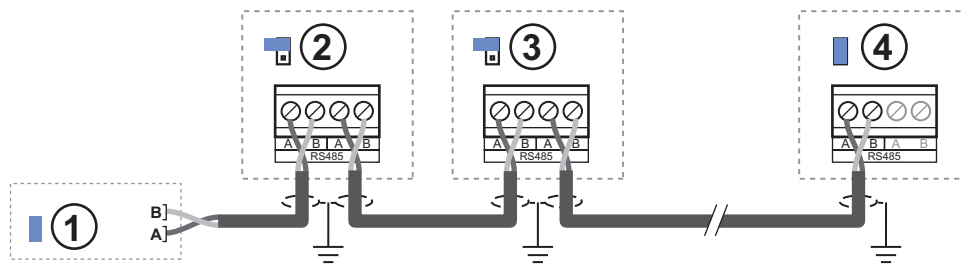
Digital output	Terminal	Selection							Description
		Alarm	Ventilation	Flap	Heating				
1	1/2 - 3								
2	1/2 - 4								
3	5/6 - 7								
4	5/6 - 8								

9.3.5 Alarm contact

The no-voltage alarm contact is interrupted in the event of an alarm or a power failure. You can include the alarm contact in an alarm circuit.



9.3.6 RS485



Legend:

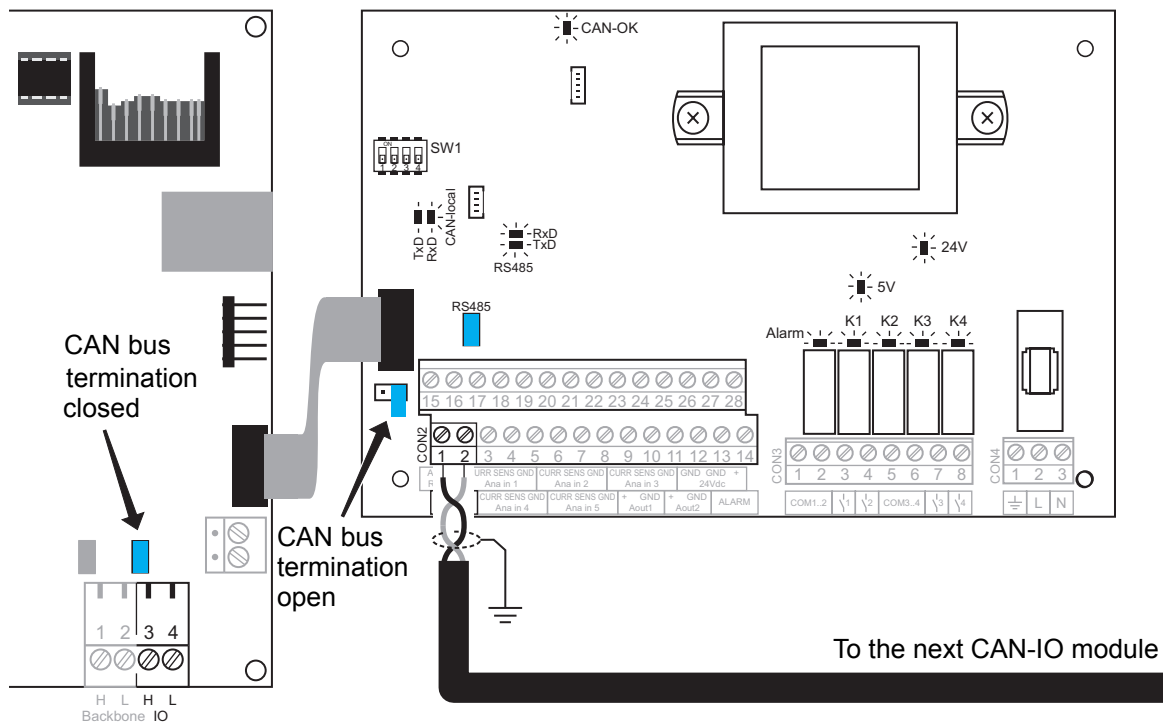
1. Module 1 - RS485 bus termination
2. Module 2 - RS485 bus termination
3. Module 3 - RS485 bus termination
4. Last module - RS485 bus termination

Note Terminate the RS485 connection with jumper JP6 at the first and last communication node.

9.4 Communication

9.4.1 CAN Local (IO modules)

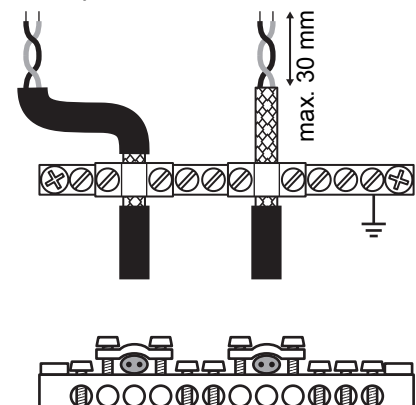
The CAN Local data bus is used for communication between the Orion-MC and the IO modules (see also the examples of connections in Appendix 2).

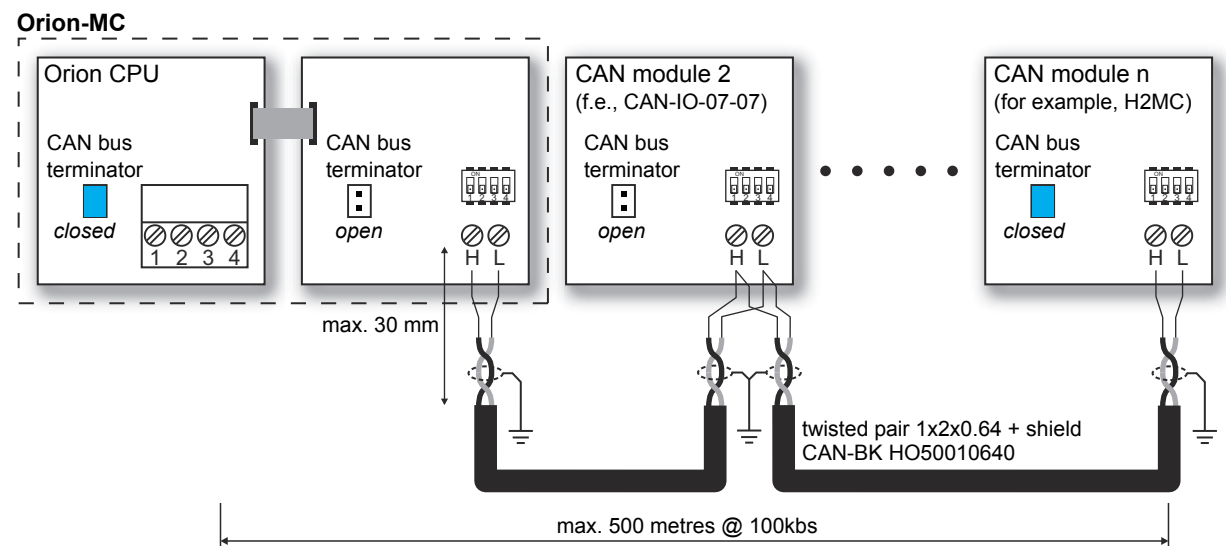


The bus must be terminated with jumpers at the first and last communication node.

Example Fit the jumpers on the CPU board and on the last IO module in the event that the CAN IO modules are connected to the integral IO module of the Orion-MC (as shown in the example above).
(Note: remove the jumper from the integral IO module or relocate it to a position where the pins are not connected.)

Example of the shield connection





Note The Orion-MC does not need to be installed at the end of the communication line.

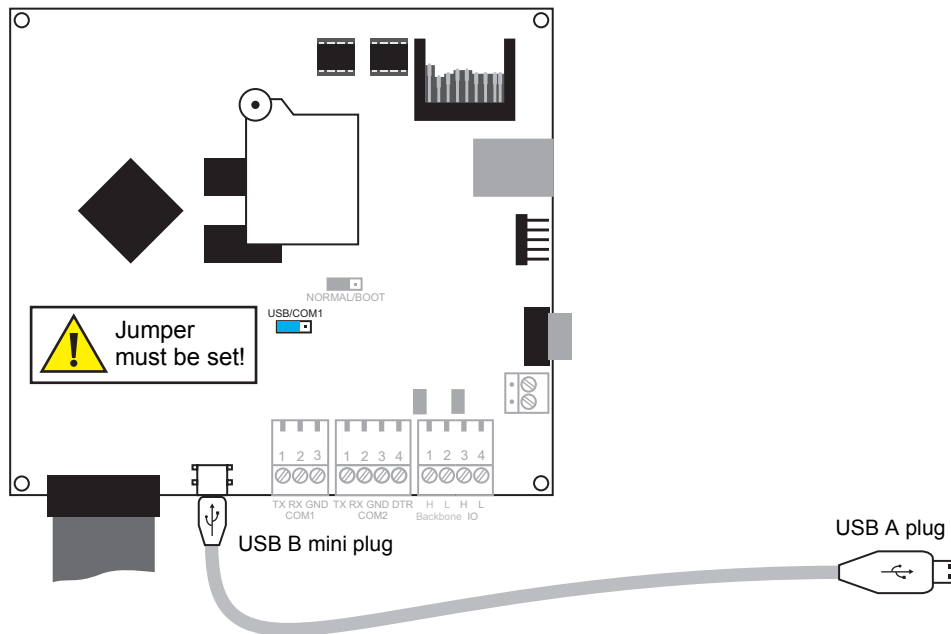
Each IO module in the communication line should be assigned a unique address. The address is configured using the IO module's DIP switch.

Address	DIP switch	Address	DIP switch	Address	DIP switch	Address	DIP switch
1		5		9		13	
2		6		10		14	
3		7		11		15	
4		8		12		16	

9.4.2 Communication with PC or Modem via USB

You can use the COM1 port or USB port (see also chapter 9.4.3). You cannot use the two ports together.

Note It is not possible to access a number of Orion-MC computers at the same time with USB.



IMPORTANT! Set the USB/COM1 jumper to USB.

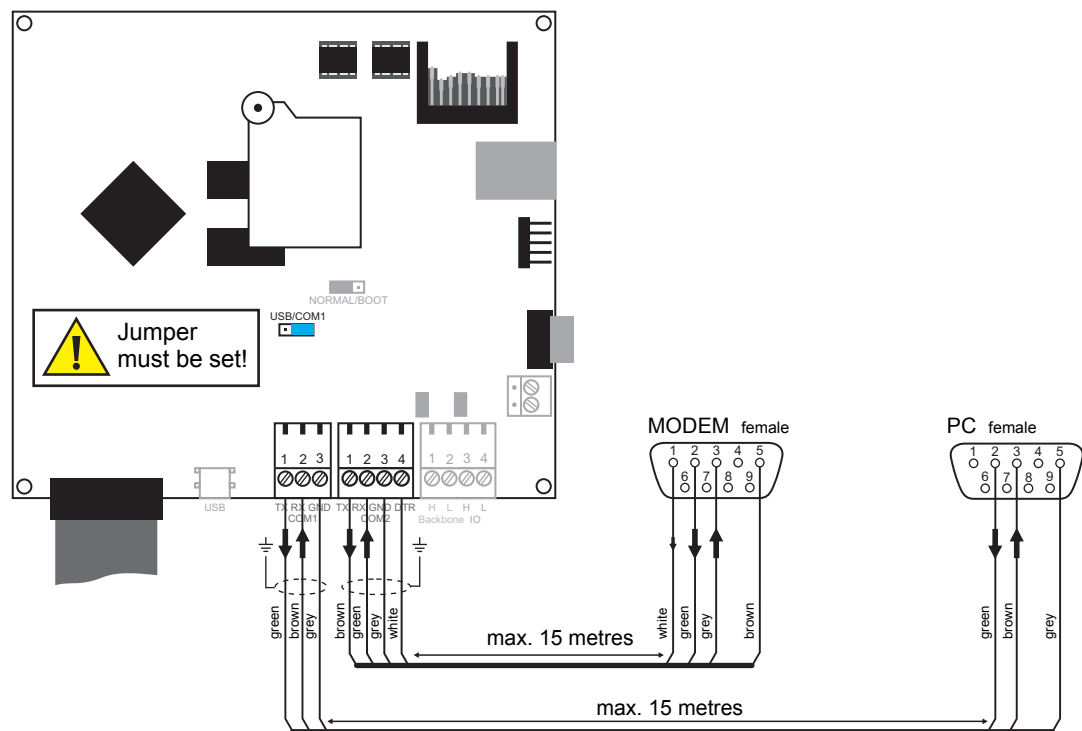
Two drivers will need to be installed for communications between the Orion-MC and a PC via USB. The general instructions for a PC with Windows are given below. Begin by installing the 'USB serial converter' driver followed by the 'USB serial port' driver.

1. Connect the Orion-MC to the PC with an USB cable.
2. Connect the Orion-MC to the power supply.
3. If Windows does not recognise the USB chip then the 'New hardware found' Wizard will start.
4. Use the Wizard to search for the drivers in the CD's 'USB' folder.
5. Install the 'USB serial converter' driver (this is selected automatically). Ignore any Windows message stating that the driver is unsigned or untested.
6. Windows will now once again find new hardware.
7. Install the 'USB serial port' driver using the same method.

9.4.3 Communication with PC or Modem via RS232

You can use the COM1 port or USB port (see also chapter 9.4.2). You cannot use the two ports together. The COM2 port can be used at any time.

Note It is not possible to access a number of Orion-MC computers at the same time with RS232.

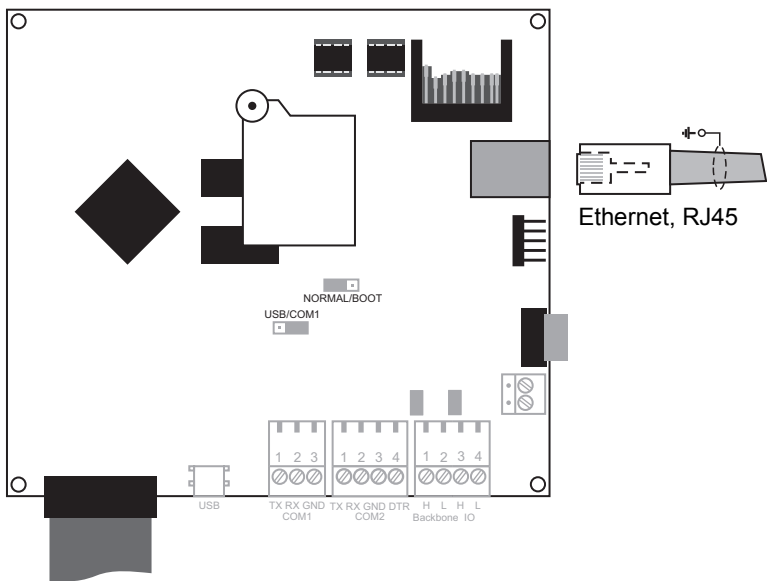


IMPORTANT! Set the USB/COM1 jumper to COM1.

Note See chapter 9.4.1 for an example of the connection of the shield.

9.4.4 Communication with PC or Modem via Ethernet

You can use Ethernet to include the Orion-MC in a local IP network.



Appendix 1 Settings lists

Appendix 1.1 Ventilation

Main group

FI

Motor groups

Group 1

36.8%

36.8%

AUTO

Settings

Default

User

Operation

Position

Fan 1

Fan 2

Fan 3

Fan ...

AUTO

0.0 %

0 %

0 %

0 %

0 %

-

%

%

%

%

%

Fan ...

Operation

Position

Offset

AUTO

0 %

0 %

-

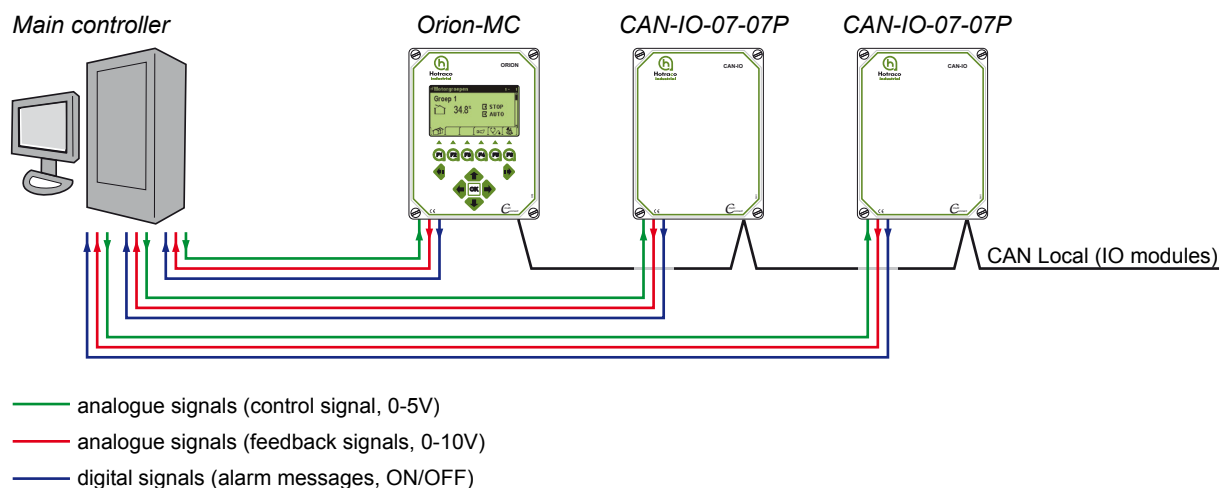
%

%

Appendix 2 System examples

Appendix 2.1 Connection to the main controller

Analog control (Orion-MC with CAN-IO-07-07P)



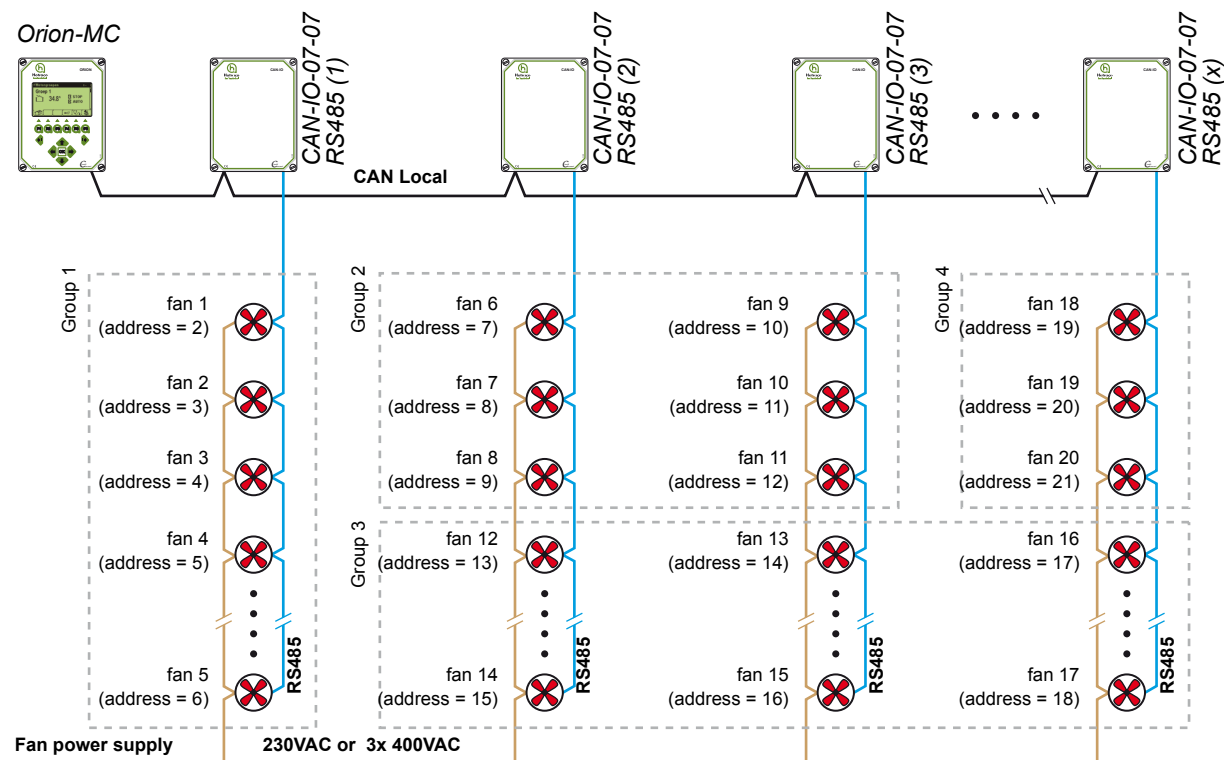
The main controller sends an analogue signal for each group to Orion-MC that corresponds with the required position. The virtual motor of the Orion-MC sends the set position as an analogue signal back to the main controller, so that the main controller receives the appropriate feedback. Digital feedback is given for alarms (the contact is de-energised in the event of an alarm).

Note

- You need one analogue input and, when relevant, one analogue output for each group: you need one digital output for each group for alarm messages.
- You can expand the number of inputs and outputs by installing extra CAN-IO-07-07P modules.

Appendix 2.2 Ventilation

IO-07-07 module with EC fans (example)



Comment: The number of fans at one RS485-Bus is limited to 31!

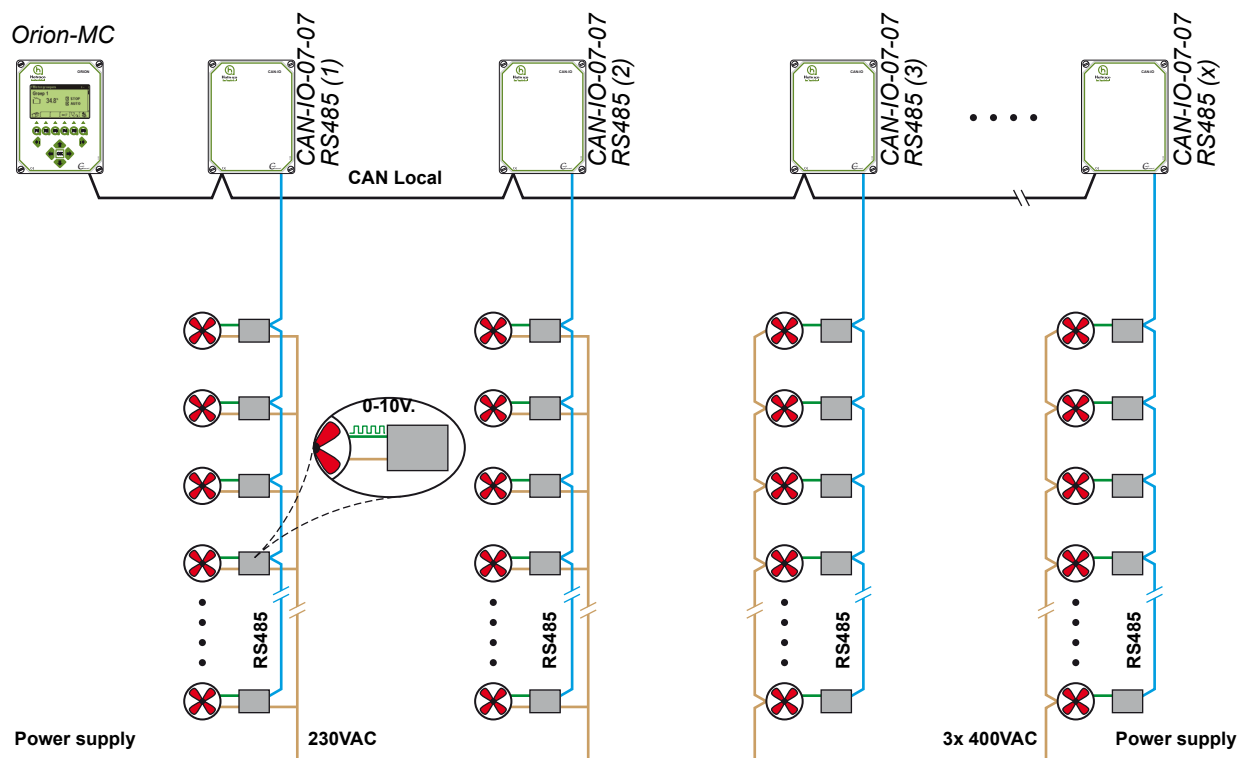
The main controller (not shown) determines the required fan position of a group. The Orion-MC ensures that the relevant fan position is set.

The required fan position is sent to the IO-07-07 modules via the CAN Local communication. The modules then convert the information into RS485 communication (ebmBus or Modbus). The EC fans then set themselves to the required speed. EC fans have an integral communication interface and separate power supplies (230V or 400V). The EC fans also send the actual fan position and any alarm messages, where relevant, to the Orion-MC.

Note

- The fans are numbered sequentially. In the above example group 1 has 5 fans (numbers 1 to 5). Group 2 has 6 fans, numbered from 6 to 11. The next fan in group 3 is then number 12, etc. The fans in a group do not all need to be connected to the same bus.
- For the purposes of communication with the Orion-MC the address must be set to the number of the fan + 1. **Example** fan number 14 is assigned the address 15 (14+1). This simplifies the replacement of a faulty fan. New fans that have yet to be configured are often set to the address 1. Once you have replaced the faulty fan you then only need to change the address of the new fan (1) to the original address of the faulty fan (see also chapter 7.4).

Example IO-07-07 module with separate EC interface (example)



Comment: Maximum 31 fans on one RS485 Bus!

The main controller (not shown) determines the required fan position of a group. The Orion-MC ensures that the relevant fan position is set.

The required fan position is sent to the IO-07-07 modules via the CAN Local communication. The module converts the information into RS485 communication and then sends it to the EC interface.

The EC interface ensures that the fan rotates at the required speed. In addition, the EC interface sends the actual fan position and any alarm messages, where relevant, to the Orion-MC.

Note

- The groups are classified in the same manner as for EC fans.
- When 400VAC fans are used then each EC interface must have a separate power supply (230VAC).

Note Maximum 31 fans on one RS485 Bus!

